Program : **Diploma in Civil and Rural Engineering**	
Course Code : **3362**	Course Title: **Construction Technology**
Semester : **3**	Credits: **3**
Course Category: **Program Core**	
Periods per week: **4 (L:3, T:1, P:0)**	Periods per semester: **60**

Course Objectives:

- To study regarding properties and testing of building materials, construction of building components
- To study properties of concrete
- To impart the basic concepts in functional requirements of building and building services

Course Prerequisites:

Topic	Course code	Course name	Semester

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	understand construction materials, properties and their uses.	15	Understanding
CO2	understand the details regarding the construction of building components	17	Understanding
CO3	Illustrate types of masonry and elements of building superstructure	13	Applying
CO4	Identify components of buildings and Suggest suitable type of foundation	13	Understanding
	Series test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1							
CO2						3	
CO3	3		3				
CO4		3					

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

On completion of the course, the student will be able to:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Understand construction materials, properties and their uses**		
M1.01	Identify classification of construction materials	3	Remembering
M1.02	Identify properties classifications and use of stone, timber	4	Understanding
M1.03	Explain manufacturing process of bricks and select ecofriendly masonry blocks	4	Understanding
M1.04	Illustrate Miscellaneous items structural steel sections for building, glasses, wood products, Industry waste materials and Specially processed products	4	Understanding

Contents:

Broad classification of materials - Natural, Artificial, special, finished and recycled. Requirements of good building stone - general characteristics of stone - Laterite stones- quarrying and dressing of stone - Structure of timber, general properties and uses of good timber - different methods of seasoning for preservation of timber - defects in timber - useof bamboo in construction. Constituents of brick earth- Manufacturing process of burnt clay brick, fly ash bricks, Autoclave Aerated Concrete (AAC) blocks – Conventional /Traditional bricks - Modular and Standard bricks, Special bricks - fly ash bricks, Characteristics of good brick, Field tests on Bricks -Classification of burnt clay bricks andtheir suitability. Structural Steel sections for buildings. Types of glass: soda lime glass, lead glass, Toughened glass and borosilicate glass and their uses. Wood products: Plywood, particle board, Veneers, laminated board and their uses. - Industrial waste materials- Fly ash, Blast furnace slag - Agro waste materials - Rice husk, Bagasse, coir fibers and their uses - Special processed construction materials- Ferro Crete, Artificial timber, MDF, HDF, PVC foam board, WPC,ACP, Artificial sand (M sand) and their uses.

CO2	**Concrete-properties, classifications and their use in construction industry**		
M2.01	Identify the properties of ingredients of concrete	6	Understanding
M2.02	Identify different grades of concrete and Ready mix Concrete	4	Understanding
M2.03	Outline properties of fresh and hardened concrete	4	Understanding
M2.04	Explain various methods of Batching, Mixing, Transportation, Placing, Compaction, Curing and surface finishing of concrete.	3	Understanding
	Series Test – I	1	

Contents:

Ingredients of cement, Manufacturing of cement- dry process, wet process, Hydration of cement, Types of cement, Physical properties of OPC & blended cement - fineness,standard consistency, setting time, soundness, compressive strength-Testing of cement: Laboratory tests-fineness, standard consistency, setting time, soundness, compressive strength

Concrete - Aggregates – Mechanical & Physical properties and tests – Grading requirements – Water quality for concrete – Admixtures – types and uses – plasticizers – accelerators – retarders –water reducing agents Making of concrete - batching – mixing – types of mixers – transportation – placing – compacting – curing Properties of concrete – fresh concrete – workability – segregation and bleeding - factors affecting workability & strength – tests on workability – tests for strength of concrete in compression, tension & flexure Special concrete - Ready mix Concrete - Fiber Reinforced Concrete - High performance Concrete - Self-compacting concrete - lightweight concrete, Geopolymer concrete-Mass concrete - Cold weather concreting - effect of cold weather on concrete - precautions to be taken while concreting in cold weather condition - Hot weather concreting - effect of hot weather on concrete - precautions to be taken while concreting in hot weather condition. Underwater concreting- Concrete exposed to sea water.

CO3	**Illustrate types of masonry and elements of building superstructure.**		
M3.01	Identify and propose types of masonry for super structure	3	Understanding
M3.02	Explain scaffolding and formwork for masonry and concreting.	5	Understanding
M3.03	Illustrate vertical communication for buildings	3	Understanding
M3.04	Identify and suggest roofing materials and methods suited for site conditions.	2	Understanding

Contents:
Types of stone masonry - Rubble masonry, Ashlar Masonry. - Pointing of stone masonryand their purpose - Precautions. Brick masonry - Terms used in brick masonry-Bonds in brick masonry- header bond, stretcher bond, English bond, Flemish bond etc. Requirements of Brick Masonry - Precautions to be observed - Hollow and Solid concrete block masonry - compositemasonry - interlock bricks. Scaffolding and Shoring - Purpose and Types - Underpinning - Formwork. Doors - Fully Paneled Doors, Partly Paneled and Glazed Doors, Flush Doors, Collapsible Doors, Rolling Shutters, Revolving Doors, Glazed Doors. Windows: Fully Paneled, Partly Paneled and Glazed, wooden, Steel, Aluminium windows,Sliding Windows, Louvered Window, Bay window, Corner window, Gable and Dormer window, Skylight - Sizes of Windows recommended by BIS. Ventilators - Fixtures and fastenings for doors and windows. Vertical Communication methods-stair, lifts and escalators. Roofs - Roofing Materials - RCC, MP Tiles, Ceramic roofing tiles, Thatched, G.I. sheets, Corrugated G.I. Sheets, Plastic and Fiber Sheets, Innovative roofing materials (Fiber reinforced cement board, Shingles etc.), Types of Roof - Flat roof, Pitched Roof - terms used in roofs - King Post truss, Queen Post Truss-terms used.

CO4	**Identify components of buildings and suggest suitable type of foundation**		
M4.01	Classify buildings as per NBC/ KMBR / KPBR and type of constructions	4	Understanding
M4.02	Identify building components and their functions	3	Understanding
M4.03	Explain construction methods for substructure and superstructure	3	Understanding
M4.04	Identify types of foundations and suggest suitable foundation for site conditions	3	Understanding
	Series Test – II	1	

Contents:

Classification of Buildings as per National Building Code/KMBR/KPBR.

Types of Constructions - Load Bearing wall Structure, Framed Structure, Composite Structure, Prefabricated structures, Special framed structure for earthquake resistance (IS 13920) - Construction methods for Structural Steel buildings.

Building Components - Functions of Building Components - Substructure - Foundation - Plinth. - DPC - Superstructure - Parts of building - Walls, Partition wall, and Cavity wall, Sill, Lintel, Doors and Windows, Floor etc.

Construction of Substructure - Job Layout - Site Clearance, Layout for Load Bearing Structure and Framed Structure by Center Line and Face Line Method.

Earthwork: Excavation for Foundation - Timbering and Strutting – Materials used for plinthFilling - Foundation - Functions of foundation - Types of foundation - Shallow Foundation and Deep Foundation - Stepped Footing, Wall Footing, Column Footing, Isolated and Combined Column Footing, Raft Foundation, Grillage Foundation, Strap footing, Pile Foundation, Well foundation and Caissons.

Text / Reference:

T/R	Book Title/Author
T1	Arora and Bindra, Building construction, Dhanpath Rai and Sons.
T2	Punmia B. C, Building construction. Laxmi Publications
T3	Rangwala S C., Engineering Materials, Charotar Publishers
T4	Shetty M.S., Concrete Technology, S. Chand & company.

Online Resources:

Sl.No	Website Link
1	https://www.youtube.com/watch?v=zmKvS_trl8I
2	https://stevenscollege.edu/civil/
3	https://www.stclaircollege.ca/programs/construction-engineering-technician